Artificial Intelligence–Based Radiomic Model in Craniopharyngiomas: A Systematic Review and Meta-Analysis on Diagnosis, Segmentation, and Classification

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■ BACKGROUND: **Craniopharyngiomas (CPs) are rare, benign brain tumors originating from Rathke's pouch remnants, typically located in the sellar/parasellar region. Accurate differentiation is crucial due to varying prognoses, with adamantinomatous CPs having higher recurrence and worse outcomes. Magnetic resonance imaging struggles with overlapping features, complicating diagnosis. This study evaluates the role of artificial intelligence (AI) in diagnosing, segmenting, and classifying CPs, emphasizing its potential to improve clinical decision-making, particularly for radiologists and neurosurgeons.**

■ METHODS: **This systematic review and meta-analysis assess AI applications in diagnosing, segmenting, and classifying on CP patients. A comprehensive search was conducted across PubMed, Scopus, Embase, and Web of Science for studies employing AI models in patients with CP. Performance metrics such as sensitivity, specificity, accuracy, and area under the curve were extracted and synthesized.**

■ RESULTS: **Eleven studies involving 1916 patients were included in the analysis. The pooled results revealed a sensitivity of 0.740 (95% confidence interval [CI]: 0.673–0.808), specificity of 0.813 (95% CI: 0.729–0.898), and accuracy of 0.746 (95% CI: 0.679–0.813). The area under the curve for diagnosis was 0.793 (95% CI: 0.719–0.866), and for classification, it was 0.899 (95% CI: 0.846–0.951). The sensitivity for segmentation was found to be 0.755 (95% CI: 0.704–0.805).**

■ CONCLUSIONS: **AI-based models show strong potential in enhancing the diagnostic accuracy and clinical decision-making process for CPs. These findings support the use of AI tools for more reliable preoperative assessment, leading to better treatment planning and patient outcomes. Further research with larger datasets is needed to optimize and validate AI applications in clinical practice.**

Key words
- Artificial intelligence
- Classification
- Craniopharyngiomas
- Deep learning
- Machine learning
- Segmentation

Abbreviations and Acronyms

ACP: Adamantinomatous craniopharyngioma
AI: Artificial intelligence
AUC: Area under the curve
CI: Confidence interval
CNN: Convolutional neural networks
CPs: Craniopharyngiomas
DL: Deep learning
DLR: Diagnostic likelihood ratio
FN: False negatives
FP: False positives
LASSO: Least absolute shrinkage and selection operator
ML: Machine learning
MRI: Magnetic resonance imaging
PCP: Papillary craniopharyngioma
PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RF: Random forest
ROC: Receiver operating characteristic
SVM: Support vector machine
TN: True negatives
TP: True positives

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INTRODUCTION

Craniopharyngiomas (CPs), representing 1.1% of nonmalignant brain and other central nervous system tumors, are rare intracranial tumors arising from epithelial remnants of Rathke's pouch, most frequently originating in the sellar/parasellar region.[1,2] CPs are represented by a slow rate of proliferation and symptoms caused by mass effect and local infiltration of the surrounding tissues.[3] CP is a benign low-grade tumor histologically with distinct phenotypes of adamantinomatous CP (ACP) and papillary CP (PCP).[2] ACPs are driven by genetic mutations in CTNNB1 and have molecularly and histologically been proposed to be of embryonic origin, while PCPs harbor BRAFV600 E mutations.[4] The differentiation of these 2 subtypes is essential because the ACP subtype is associated with higher rates of recurrence and poor prognosis compared to PCPs, largely due to its infiltrative nature causing difficulties in radical resection.[5] The differential diagnosis may be a challenge when the radiological features overlap other lesions of the sellar and parasellar region. In cases of purely cystic CPs, the main diagnostic challenges in children and adults are Rathke's cleft cysts, arachnoid cysts, dermoid cysts, and epidermoid cysts.[5]

Artificial intelligence (AI) comprises several technologies that computationally simulate human intelligence. Machine learning (ML) is part of AI that generates predictions using mathematical algorithms. Deep learning (DL), inspired by the neurological architecture of the brain, is a subgroup of ML that makes predictions using multilayered neural network algorithms.[6] Deep neural network is a type of DL that has been used as an elemental approach in image preprocessing and improving the reliability of diagnostic results by exploiting big data in medical imaging.[7-10]

Compared to other types of brain tumors, segmentation of CPs using deep neural network is more challenging due to the considerable variability of ACP and PCP in their imaging presentation. PCP usually presents as a noncalcified and solid entity. At the same time, ACPs demonstrate calcifications, being cystic, and contrast take-up in the cyst walls in almost 90% of the cases.[7] Inconsistent, low amounts of Dice coefficient have been reported among literature for various brain tumors such as the pituitary and suprasellar cistern, due to small tissue and vulnerability to compression in a small area.[11] Radiographic diagnosis offers a noninvasive solution. However, the accuracy of clinical expertise in diagnosing pediatric suprasellar brain tumors, such as ACP, from preoperative magnetic resonance imaging (MRI) is around 86%, yet lower than neurosurgical biopsy (91%).[12] Understanding of radiographs can be limited by the low spatial resolution of radiology images and atypical visual features that make diseases look similar. DL can detect brain tumors from radiology images and is sensitive to minor image alterations that are not observable to humans.[12] On the other hand, radiomics-based ML approaches are increasingly evaluated in the field of neuro-oncology, since they can provide valuable quantitative information reflecting the intratumoral heterogeneity, which is less reliably interpreted by human expert evaluation, via high-throughput extraction of imaging features.[13]

This systematic review and meta-analysis particularly focus on the role of AI in diagnosing CPs to give a comprehensive view of the current status of AI in diagnosing and differentiating CPs. AI can be utilized to improve the reliability of radiographic diagnosis by supporting the clinician when uncertain. By running a systematic review, we gathered all the relevant data in the field to bookmark the place we are standing in the AI-based diagnosis of CPs.

METHOD

This study is an International Prospective Register of Systematic Review (PROSPERO ID: CRD42024627598) that systematically reviews and executes a meta-analysis of research using different ML algorithms to anticipate AI application in the diagnosis, segmentation, and classification of CPs. It follows the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).[14]

Search Strategy

We searched PubMed, Embase, Scopus, and Web of Science databases to identify studies regarding ML-based models in CP diagnosis, segmentation, and classification. To ensure no relevant articles were missed, the first 100 results from Google Scholar were reviewed as a supplementary search on December 14, 2024. We provide an unrivaled search strategy for individual databases with the following search terms: ("Craniopharyngioma" OR "Adamantinomatous" OR "Neoplasm" OR "Rathke Cleft") AND ("Artificial intelligence" OR "Machine learning" OR "Deep learning"). Moreover, other similar keywords have been evaluated. There was no constraint about publication date, language, or publication type.

Inclusion Criteria

1)English language; 2) studies involving human subjects diagnosed with CPs, irrespective of age or gender; 3) studies that directly address the use of AI, ML, or DL for forecasting, segmentation, or classification of CPs; 4) studies reporting accuracy, sensitivity, specificity, precision, recall, F1 score, area under the curve (AUC), or other diagnostic and performance metrics for AI-based models; 5) studies providing sufficient data to evaluate the predictive, segmentation, or classification performance of the AI models, including metrics like true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN).

Exclusion Criteria

1)Nonoriginal articles and case reports; 2) non-English studies; 3) studies not addressing AI, ML, or DL applications specific to CPs, such as those focusing on general brain tumor analysis or non-AI-based approaches; 4) studies involving conditions unrelated to CPs or experimental models not involving human data (e.g., animal studies); 5) studies lacking detailed diagnostic or performance metrics (e.g., sensitivity, specificity, accuracy) or those with missing data, preventing a comprehensive analysis; 6) studies with unclear methodologies, insufficient sample sizes, high risk of bias, or failure to meet established quality assessment criteria.

Study Selection

EndNote version 21 has been used to select retrieved articles. Two reviewers (N.F. and N.F.) independently screened studies according to the inclusion and exclusion criteria. With the cooperation of a third reviewer (I.M.), the most relevant studies were chosen following the title and abstract assessment screening steps. The extraction stage contained the studies that comprise the eligibility criteria.

Data Extraction

Two reviewers (N.F. and N.F.) separately extracted the intended data from included articles, including study characteristics first name, year of publication, country, design of the study, number of patients, mean age, gender composition, type of reference sequence computed tomography or MRI or DSA. Consequently, performance comparisons of ML algorithms including accuracy, sensitivity, specificity, F1 score, and AUC were extracted. Using a predesigned Excel sheet, the information above has been elicited.

Quality Assessment

The Prediction model Risk Of Bias Assessment Tool (PROBAST)[15] tool was used to check the risk of bias and how well the studies matched the purpose of the research. This tool looks at 4 areas:

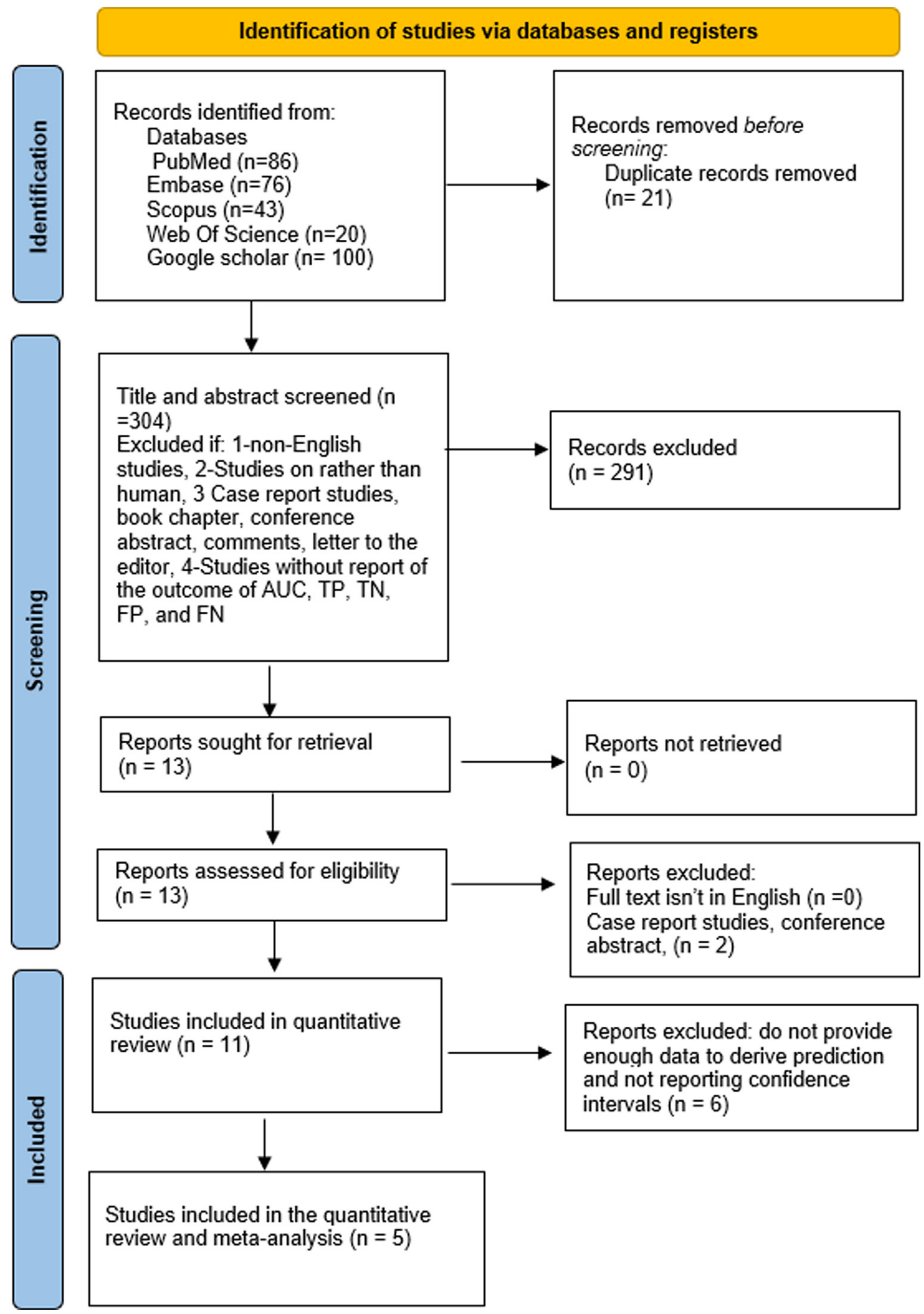

Figure 1. PRISMA flowchart of the study selection process. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; TP, true positives; FP, false positives; TN, true negatives; FN, false negatives; AUC, area under the curve.

analysis, outcomes, predictors, and participants. Each area was rated as having a low, high, or unclear risk of bias. The prediction models were also checked to ensure they fit the study's goals. In addition to PROBAST, we also evaluated the methodological quality of each study using the Newcastle-Ottawa Scale,[16] scoring studies across 3

Table 1. Demographic and characteristics of inclusion studies

Author/ Year	Type of Study	Country	No. of Extracted/ Final (Radiomics Features)	No of Patients in Train/Test/Validation Group	Location of Tumor	Total Mean Age/ Female %	Inclusion Criteria	Exclusion Criteria	Type of Treatment	Grade of Tumor	Type of Outcome/ Definition
C Chen et al./ 2022[7]	Retrospective	China	NA	212/52/38	Suprasellar and intrasellar.	32.92/ 40.7	Pathologically confirmed craniopharyngiomas, presurgical MR scans, clear imaging without severe motion artifacts.	Unreadable MRIs with severe motion artifacts, tumor intervention history prior to MR scans, such as biopsy and radiotherapy, unclear pathological subtypes	NA	NA	Segmentation/ segmentation refers to the process of delineating craniopharyngioma boundaries on MRIs. It is crucial for tumor identification, volumetric assessment, and therapy planning. Automated segmentation using the modified U-Net model improves accuracy and reduces the workload compared to manual methods.
E Prince et al./ 2022[*]	Retrospective	USA	NA	53/33	Sellar/ Suprasellar region.	NA	Patient with histologically confirmed ACP or NOTACP lesion with preoperative CT and MRI imaging available	NA	NA	grade I	Classification/radiographic feature of adamantinomatous craniopharyngioma (ACP), often observed in the sellar/suprasellar region. It is highlighted as one of the distinguishing imaging characteristics of ACP, along with heterogeneous solid and cystic regions. This feature is essential for noninvasive diagnosis and supports the performance of machine learning models in differentiating ACP from other lesions.
E Prince et al./ 2023[12]	Retrospective	USA	2048	53/33	Suprasellar	NA	Preoperative MRI data with histopathologically confirmed ACP or NOTACP tumors.	NA	NA	NA	Classification

Z. Huang et al./ 2021[13]	Retrospective	China	1213/NA	99/65	Suprasellar and intrasellar/ suprasellar.	28.22/ 35.36	Pathologically confirmed craniopharyngioma	Unclear pathological subtypes; missing preoperative MRI sequences: axial T1-w, T2-w, and CET1-w; unreadable MRI data because of severe artifacts, history of any major intervention prior to MR scans, such as radiotherapy and biopsy	Surgery	Benign (subtypes: ACP and PCP).	Classification/calcification is mentioned as an imaging characteristic aiding subtype differentiation. Instead, it is considered part of the overall radiomic features used for classification.
X Yan et al./ 2023[11]	Retrospective	China	66	106/27	NA	30.8/ 42.1	Diagnosed with craniopharyngioma, available preoperative contrast-enhanced sagittal MRI	Poor imaging quality or artifacts, missing clinical data.	NA	Benign (classified into Q, S, T subtypes).	Segmentation/ automatically identifying and delineating craniopharyngiomas and adjacent structures on MRI using a deep learning model. The model achieved a high Dice coefficient of 0.951 for tumors, enabling precise preoperative planning and QST classification to improve surgical outcomes.
X. Chen et al./ 2019[18]	Retrospective	China	1021/9	44.00	Sellar region (parasellar tumors)	43.9/ 34.1	Patients with histologically confirmed craniopharyngiomas and preoperative MRI data	Incomplete MRI data or lack of pathological confirmation	Surgery	WHO I	Diagnosis/MRI-based radiomics model for noninvasive diagnosis of craniopharyngiomas, predicting pathological subtypes and genetic mutations (BRAF V600 E and CTNNB1).
Y. Teng et al./ 2022[4]	Retrospective	China	NA	97	Sellar-suprasellar region.	47.15/ 45	Histopathological confirmation of ACP or PCP, preoperative MRIs available, adult patients (over 18 years).	Poor image quality (motion artifacts), prior treatments (e.g., radiotherapy or biopsy).	Surgery	Benign (ACP and PCP subtypes).	Diagnosis/CNN models, including ResNet50, to differentiate ACP from PCP on contrast-enhanced T1-weighted MRIs

No., number; NA, not available; ACP, adamantinomatous craniopharyngioma; PCP, papillary craniopharyngioma; MRI, magnetic resonance imaging; CT, computed tomography; T1-w: T1-weighted imaging; T2-w, T2-weighted imaging; CE-T1-w, contrast-enhanced T1-weighted imaging; CNN, convolutional neural network; ANN, artificial neural network; U-Net, U-Net deep learning model; ResNet50, Residual Neural Network 50; CTNNB1, catenin β-1; BRAF, BRAF V600 E; WHO, World Health Organization.

*Prince EW, Whelan R, Mirsky DM, et al. Robust deep learning classification of adamantinomatous craniopharyngioma from limited preoperative radiographic images. Scientific reports 2020; 10 (1):16885.

†Zhao Z, Xiao D, Nie C, et al. Development of a nomogram based on preoperative bi-parametric MRI and blood indices for the differentiation between cystic-solid pituitary adenoma and craniopharyngioma. Frontiers in Oncology 2021; 11:709321.

‡Zhang Y, Shang L, Chen C, et al. Machine-learning classifiers in discrimination of lesions located in the anterior skull base. Front Oncol. 2020; 10: 752. 2020.

Continues

Table 1. Continued

Author/ Year	Type of Study	Country	No. of Extracted/ Final (Radiomics Features)	No of Patients in Train/Test/Validation Group	Location of Tumor	Total Mean Age/ Female %	Inclusion Criteria	Exclusion Criteria	Type of Treatment	Grade of Tumor	Type of Outcome/ Definition
B Chen et al./ 2022[19]	Retrospective	China	NA/40	86/21	Anterior skull base.	22/ 49.05	Pathologic confirmation of germinoma or CP, available high-quality preoperative, MR scans performed in the radiological department, the lesion was located in the anterior skull base	Incomplete medical records in diagnosis or treatment, recorded history of any other intracranial disease, patients had undertaken a treatment, such as surgery, radiotherapy or chemotherapy prior to the available MR scan	Surgery	NA	Diagnosis/differentiation among three types of sellar-suprasellar masses: pituitary adenoma, craniopharyngioma, and Rathke's cleft cyst. An artificial neural network (ANN) was employed to aid radiologists in diagnosing these lesions based on MRI findings and patient data
C Jiang et al./ 2023[20]	Retrospective	China	1037	390	Sellar-suprasellar	41.6/ 63.8	Adults with intraoperatively and histopathologically confirmed Rathke's cleft cysts, cystic craniopharyngiomas, cystic pituitary adenomas, or pituitary apoplexy; preoperative contrast-enhanced T1 (T1—CE)-weighted and T2-weighted MRI	Those with MRI scans conducted at external institutions	NA	NA	Diagnosis
Zhao et al./ 2020†	Retrospective	China	1015 extracted, 18 final	182/90/NA	Sellar/ parasellar region	50.11/ 47.8	Pathologically confirmed CS-PA or CP; preop CE—T1 and T2 MRI; $\geq$3 lesion slices; blood tests (routine + liver) within 2 weeks; no infection	Incomplete MRI; history of cranial trauma/tumor/ surgery/hematologic or active infection; prior chemo/radio/hormone therapy	Surgery	Benign (WHO grade I)	Diagnosis
Zhang et al./ 2020‡	Retrospective	China	40 extracted; final feature count varies by selection method	188/47/NA	Anterior skull base (sellar, parasellar, intrasellar)	43.6/ 55.3	Pathologically confirmed pituitary adenoma, meningioma, craniopharyngioma, or Rathke cleft cyst; anterior skull base location; preoperative MRI available	History of other intracranial disease (e.g. stroke, infection); prior antitumor treatment before MRI; incomplete records	Surgery	Benign (WHO grade I)	Diagnosis

No., number; NA, not available; ACP, adamantinomatous craniopharyngioma; PCP, papillary craniopharyngioma; MRI, magnetic resonance imaging; CT, computed tomography; T1-w: T1-weighted imaging; T2-w, T2-weighted imaging; CE-T1-w, contrast-enhanced T1-weighted imaging; CNN, convolutional neural network; ANN, artificial neural network; U-Net, U-Net deep learning model; ResNet50, Residual Neural Network 50; CTNNB1, catenin β-1; BRAF, BRAF V600 E; WHO, World Health Organization.

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‡Zhang Y, Shang L, Chen C, et al. Machine-learning classifiers in discrimination of lesions located in the anterior skull base. Front Oncol. 2020; 10: 752. 2020.

Figure 2. Frequency of algorithms used in the included studies. CNN, convolutional neural network; GA, genetic algorithm; XGBoost, extreme gradient boosting; RF, random forest; ResNet, Residual Neural Network; LASSO, least absolute shrinkage and selection operator; SVM, support vector machine; SBM, statistical brain mapping; LDA, linear discriminant analysis; DC, decision classifier; LR, logistic regression; KNN, k-nearest neighbors; GBDT, gradient boosting decision tree; DT, decision tree.

domains: selection, comparability, and outcome, to provide a complementary assessment of observational study rigor.

Statistical Analysis

A meta-analysis was performed using Stata version 17 to synthesize diagnostic performance metrics, including the AUC, sensitivity, specificity, and accuracy. A random-effects model was employed to account for interstudy variability. The primary objective was to calculate the overall AUC by aggregating individual AUC values along with their respective confidence intervals (CIs) (upper and lower bounds) and to assess potential differences across studies.

In addition, we conducted a bivariate meta-analysis of diagnostic accuracy using the MIDAS package in Stata. TP, FP, TN, and FN were calculated from each algorithm's sensitivity and specificity in each study. These values were then used to derive pooled estimates of sensitivity, specificity, positive and negative diagnostic likelihood ratios (DLR), diagnostic odds ratio, diagnostic score, and AUC under a hierarchical random-effects framework.

Heterogeneity (variance) between studies was assessed using the I^2 statistic. According to Higgins and Thompson's classification, I^2 values of up to 25% were categorized as low heterogeneity, while values of 50% and 75% represented moderate and high heterogeneity, respectively.[17] Additionally, a leave-one-out sensitivity analysis was conducted to evaluate the robustness of the results. This analysis involved sequentially excluding each study to examine its influence on the overall diagnostic performance metrics and to ensure the stability and reliability of the findings.

RESULT

Study Selection Process

Our systematic search from this study in the electronic databases collected 325

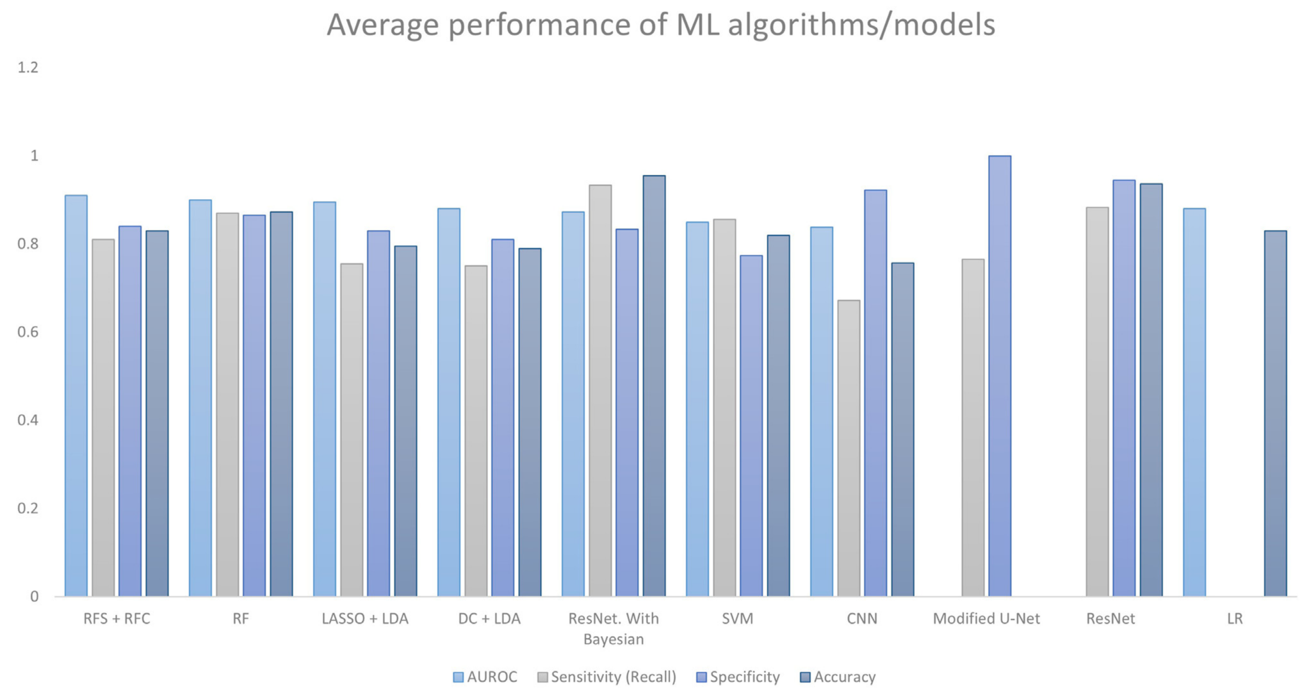

Figure 3. Performance comparison of AI algorithms, including accuracy, sensitivity, specificity, and AUC. ML, machine learning; RF, random forest; LDA, linear discriminant analysis; LASSO, least absolute shrinkage and selection operator; ResNet, Residual Neural Network; SVM, support vector machine; CNN, convolutional neural network; RFC, random forest classifier; DC, decision classifier; LR, logistic regression.

records. Twenty-one duplicate articles were removed, and 304 studies underwent the first screening step according to the eligibility criteria; 13 articles met the criteria. These 13 articles underwent full-text screening. Subsequently, 11 of them met the eligibility criteria and were included in the data extraction; all screening steps are illustrated in the PRISMA flowchart in **Figure 1**.

Study Characteristics

Eleven studies, including a total of 1916 patients, with sample sizes ranging from 44 to 302 participants, were analyzed in our study. In terms of gender distribution, the proportion of female/male patients was percentages of 46.3%. Among the 11 studies reviewed, 8 (72.72%) were conducted in China, while 2 (18.18%) took place in the United States. All studies were retrospective. The average age of the patients included was 27.1 years. MRI was also the most commonly used imaging modality among studies, and radiomic features were also commonly used as input variables in most studies (**Table 1**).

Machine Learning-Based Models Characteristics

Among our included studies, various ML and DL models were utilized for data analysis. The majority of the applied methods (78.6%) were ML-based, while the 3 algorithms most frequently utilized were including random forest (RF), support vector machine (SVM), and least absolute shrinkage and selection operator (LASSO). Additionally, feature selection techniques such as SelectKBest and LASSO and an optimization approach genetic algorithm were employed. On the other hand, DL models comprised 21.4% of the applied methods, including convolutional neural networks (CNN), ResNet, and a modified version of U-Net. (Random forest classifier [RFC] + RFC) and RF show the high performance (AUC: 0.91 and 0.89, respectively) between other algorithms/models (**Figures 2** and **3**) (**Table 2**).

Meta-Analysis

Segmentation. The pooled sensitivity for segmentation of CPs was 0.755 (95% CI: 0.704–0.805) under a random-effects model, with negligible heterogeneity ($\tau^2 = 0.0000$; $I^2 = 0.0\%$; $P = 0.5914$), and the result was statistically significant ($z = 29.33$; $P < 0.0001$) (**Figure 4**).

Classification. The pooled AUC for classification of CPs was 0.899 (95% CI: 0.846–0.951) under a random-effects model, with negligible heterogeneity ($\tau^2 = 0.0000$; $I^2 = 0.0\%$; $P = 0.6156$), and the overall effect was statistically significant ($z = 33.46$; $P < 0.0001$) (**Figure 5**).

Diagnose. Pooled sensitivity and specificity were 0.87 (95% CI: 0.77–0.93) and 0.8 (95% CI: 0.72–0.86) with negligible ($I^2 = 0\%$ and 0%) heterogeneity, respectively. The positive DLR was 4.26, and the negative DLR was 0.16, corresponding to no significant and very low heterogeneity (I^2 0% and 0%), respectively. The diagnostic score was 3.27 and the diagnostic odds ratio was 26.33 ($I^2 = 96.74\%$ and 100%, respectively). The pooled area under the summary receiver operating characteristic curve had an AUC of 0.91 (95%

Table 2. ML models characteristics and performance metrics

Author/ Year	Validation	Type of Reference	Input Characteristics	Selected Features	Method of Radiomics	Sequences Analyzed	Best Sequences Analyzed	ML Algorithms and Models	Best Predictor	Accuracy	Sensitivity	Specificity	AUC
C Chen et al./ 2022[7]	Internal validation (8:2 split) and external testing	MR	Radiomics	Tumor boundary, solid/cystic composition, relationship with surrounding structures	Semiautomated	T1-weighted imaging, T2-weighted imaging, contrast-enhanced T1 images (T1CE), fluid attenuated inversion recovery	Contrast-enhanced T1 images (T1CE)	CNN, Modified U-Net	CNN	NA	0.736 (0.651—0.822)	0.9996 (0.9994—0.9996)	NA
E Prince et al./ 2022[*]	5-fold cross-validation and hold-out validation	CT, MRI, combined CT-MRI dataset	Radiomics	NA	Semiautomated	T1-weighted MRI,non contrast CT	Combined CT-MRI	CNN, GA	ResNet variant	NA	NA	NA	87.8
E Prince et al./ 2023[12]	NA	MRI	Radiomics	NA	Semiautomated	T1-weighted MRI	T1-weighted MRI	XGBoost, RF, ResNet. With Bayesian	ResNet with Bayesian	0.955	0.933	0.833	0.871
Z. Huang et al./ 2021[13]	Internal validation - external validation	MRI	Radiomics	7 images features	Semiautomated	Axial (T1-w),T2-w, (CET1-w) on 3.0 T or 1.5 T magnetic resonance scanners	Combined multiparametric MRI sequences	SelectKBest, LASSO, SVM, SelectKBest (SVM-RFE)	SVM	0.869	0.909	0.727	0.899 (0.833—0.965)

NA, not available; MR, magnetic resonance; CT, computed tomography; MRI, magnetic resonance imaging; T1CE, contrast-enhanced T1 images; CNN, convolutional neural network; GA, genetic algorithm; XGBoost, extreme gradient boosting; RF, random forest; ResNet, Residual Neural Network; Bayesian, Bayesian inference model; LASSO, least absolute shrinkage and selection operator; SVM, support vector machine; SVM-RFE, support vector machine with recursive feature elimination; nnUNet, no-new-net U-Net model, CET1-w, contrast-enhanced T1-weighted imaging; MPRAGE, magnetization-prepared rapid gradient-echo; SBM, statistical brain mapping; RFC, random forest classifier; LDA, linear discriminant analysis; DC, decision classifier; LR, logistic regression; GLSZM, gray level size zone matrix; GLDM, gray level dependence matrix; NGTDM, neighborhood gray tone difference matrix; KNN, k-nearest neighbors; GBDT, gradient boosting decision tree; DT, decision tree; GLCM, gray-level co-occurrence matrix; GLRLM, gray-level run-length matrix; T1WI, T1-weighted imaging; T2WI, T2-weighted imaging.

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Continues

Table 2. Continued

Author/ Year	Validation	Type of Reference	Input Characteristics	Selected Features	Method of Radiomics	Sequences Analyzed	Best Sequences Analyzed	ML Algorithms and Models	Best Predictor	Accuracy	Sensitivity	Specificity	AUC
X Yan et al./ 2023[11]	5-fold cross-validation	MRI	Multimodal radiomics features derived from automatic segmentation.	Tumor volume, diameters and relative location, tumor position relative to the brain, pituitary volume, ventricular dilation	Semiautomated	CET1-w	Sagittal CET1-w	nnUNet, ResNet	ResNet (Q, S, T, 3Type)	0.951	0.9656	0.9519	NA
X. Chen et al./ 2019[18]	10-fold cross-validation	MRI	Clinical, genomic	Pathological subtypes, BRAF and CTNNB1 mutational status	Semiautomated	T1-MPRAGE MR images using a Magnetom Trio 3 T (Siemens) scanner	T1-weighted MPRAGE	RF	RF (pathological types, BRAF gene, CTNNB1 gene)	0.91	0.92	0.89	0.96
Y. Teng et al./ 2022[4]	5-fold cross-validation	MRI	Clinical and radiomics features	Age, sex, onset duration, symptoms, hypothalamic involvement, location, tissue structure, shape, maximum diameter	Automated	Coronal contrast-enhanced T1-weighted MRIsT	Contrast-enhanced T1-weighted MRIs	SBM, RF, CNN	CNN	0.757 0.	0.608	0.845	0.838
B Chen et al./ 2022[19]	4:1 split for training and validation cohorts, repeated for 100 cycles.	MRI	Radiomics	Histogram-based features, shape-based features, texture features (GLCM, GLRLM, GLZLM, NGLDM).	Semiautomated	Contrast-enhanced T1WI, T2WI.	Both contrast-enhanced T1WI and T2WI.	LASSO, RFC, SVM, LDA, DC	Combined (RFS + RFC)	0.83	0.81	0.84	0.91

C Jiang et al./ 2023[20]	5-fold cross-validation	Contrast-enhanced T1-wieghted MTI, T2-weighted MRI	Radiomics	Texture features (GLCM, GLRLM, GLSZM, GLDM, NGTDM), first-order statistics, and shape-based features.	Semiautomated	T1-weighted contrast-enhanced, T2-weighted MRI	T2-weighted MR	LR, SVM, AdaBoost, RF,	SVM	75.32	NA	NA	0.8534
Zhao et al./ 2021†	5-fold cross-validation	MRI (contrast-enhanced T1 and T2)	Radiomics features from MRI and clinical blood indices	Eighteen radiomic features after univariate *t*-test filtering and PCA dimensionality reduction	Fully automated	CE-T1 and T2-weighted MRI	Combined CE-T1 + T2 outperformed single-sequence	LR	LR	0.83			0.88
Zhang et al./ 2020‡	NA	MRI (contrast-enhanced T1)	Radiomics features from MRI and clinical features (age, gender)	Subset of 40 radiomic features selected via LASSO regression	Manual ROI segmentation; automated feature extraction with LIFEx software	Contrast-enhanced T1-weighted MRI	Single CE-T1 (no sequence comparison)	LASSO, LDA, RF, SVM, Adaboost, KNN, GaussianNB, LR, GBDT and DT (ML)	LASSO feature selection + Linear discriminant analysis	0.8	0.888	0.734	0.804

NA, not available; MR, magnetic resonance; CT, computed tomography; MRI, magnetic resonance imaging; T1CE, contrast-enhanced T1 images; CNN, convolutional neural network; GA, genetic algorithm; XGBoost, extreme gradient boosting; RF, random forest; ResNet, Residual Neural Network; Bayesian, Bayesian inference model; LASSO, least absolute shrinkage and selection operator; SVM, support vector machine; SVM-RFE, support vector machine with recursive feature elimination; nnUNet, no-new-net U-Net model, CET1-w, contrast-enhanced T1-weighted imaging; MPRAGE, magnetization-prepared rapid gradient-echo; SBM, statistical brain mapping; RFC, random forest classifier; LDA, linear discriminant analysis; DC, decision classifier; LR, logistic regression; GLSZM, gray level size zone matrix; GLDM, gray level dependence matrix; NGTDM, neighborhood gray tone difference matrix; KNN, k-nearest neighbors; GBDT, gradient boosting decision tree; DT, decision tree; GLCM, gray-level co-occurrence matrix; GLRLM, gray-level run-length matrix; T1WI, T1-weighted imaging; T2WI, T2-weighted imaging.

*Prince EW, Whelan R, Mirsky DM, et al. Robust deep learning classification of adamantinomatous craniopharyngioma from limited preoperative radiographic images. Scientific reports 2020; 10 (1):16,885.

†Zhao Z, Xiao D, Nie C, et al. Development of a nomogram based on preoperative bi-parametric MRI and blood indices for the differentiation between cystic-solid pituitary adenoma and craniopharyngioma. Frontiers in Oncology 2021; 11:709321.

‡Zhang Y, Shang L, Chen C, et al. Machine-learning classifiers in discrimination of lesions located in the anterior skull base. Front Oncol. 2020; 10: 752. 2020.

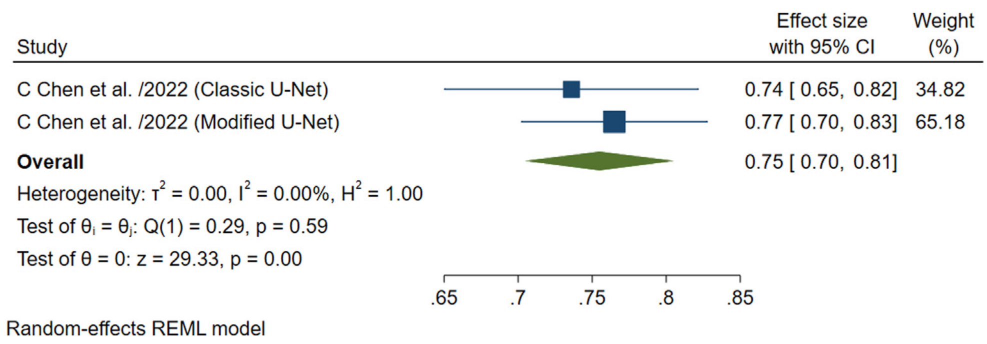

Figure 4. Forest plot of sensitivity meta-analysis for segmentation.

CI: 0.88–0.93) (Supplementary File. Supplementary Figures S1-S4).

Sensitivity Analysis. For segmentation, the sensitivity analysis showed that the pooled sensitivity remained stable across different studies, with values of 0.77 (95% CI: 0.70–0.83) and 0.74 (95% CI: 0.65–0.82), both with statistically significant results ($P = 0.000$). For classification, the exclusion of individual studies did not significantly affect the pooled AUC, which remained in the range of 0.90 to 0.91 (95% CI), indicating consistent performance across various validation phases. In addition, the analysis for accuracy showed a stable effect size of 0.74 (95% CI: 0.65–0.83) when omitting the study by Y. Teng et al. (2022) for CNN and SVM models, and 0.76 (95% CI: 0.67–0.84) for the RF model, all with statistically significant results ($P = 0.000$). These findings demonstrate that the models for segmentation, classification, and diagnostic accuracy show stable and consistent performance, regardless of the omission of individual studies (Supplementary Figures S5-S6).

Publication Bias. The trim-and-fill method for segmentation suggested a small positive bias, with a slight increase in effect size from 0.755 to 0.765. The funnel plot for segmentation was symmetric, indicating minimal impact on the result. For classification, Egger's test showed no significant bias ($P = 0.4875$), and the funnel plot also displayed symmetry (Supplementary Figures S7-S8).

Quality Assessment

Quality Assessment (PROBAST). Across the 4 domains of participants, predictors, outcomes, and analysis, 83.33% of studies demonstrated low risk of bias. In 16.67% of studies, the risk of bias was unclear, particularly in the predictor and outcomes domains. No studies were rated as having a high risk of bias (Figure 6).

Applicability of the ML Tools (PROBAST). For applicability, 83.33% of studies showed low concern in the participants and outcome domains, while 66.67% of studies demonstrated low concern in the predictor domain. In 33.33% of studies, the applicability of predictors was unclear. No domains were noted as high concern among them (Figure 7).

Newcastle-Ottawa Scale

The quality of the 11 studies included in this review was assessed using the Newcastle-Ottawa Scale across 3 domains: selection, comparability, and outcome. All studies were retrospective with well-defined participant selection, though one study lacked sufficient detail. In terms of comparability, 82% of studies employed appropriate validation methods (e.g., 5-fold cross-validation), while 2 studies did not report validation, affecting their scores. Regarding outcomes, 91% of studies provided reliable diagnostic

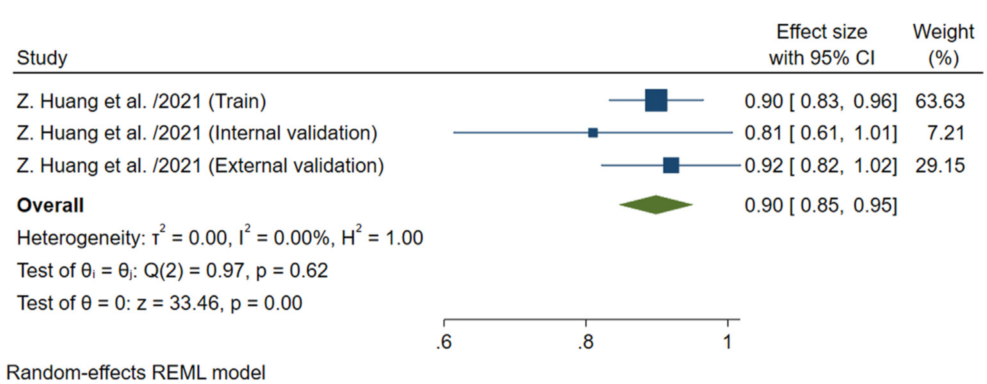

Figure 5. Forest plot of AUC meta-analysis for classification. AUC, area under the curve.

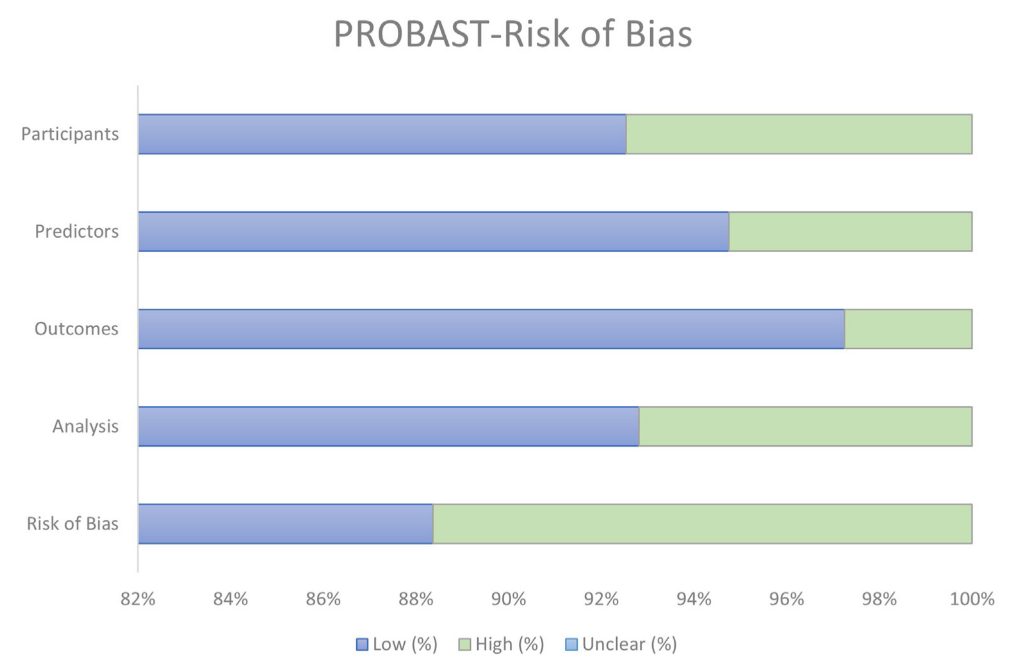

Figure 6. Prediction model Risk Of Bias Assessment Tool chart showing the distribution of risk of bias across the domains of participants, predictors, outcome, and analysis.

metrics such as accuracy, sensitivity, and specificity, with one study having incomplete reporting. Overall, the studies demonstrated high quality, although transparency in reporting validation and outcomes could be improved.

DISCUSSION

PCP and ACP represent unique characteristics through the MRI, differing in the

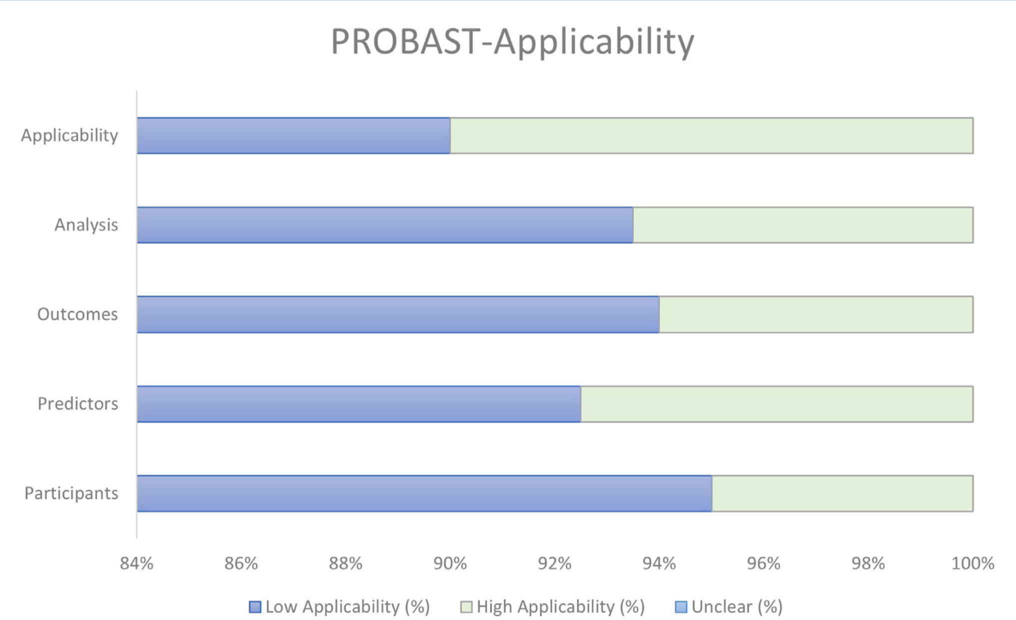

Figure 7. Prediction model Risk Of Bias Assessment Tool chart illustrating the applicability assessment across the domains of participants, predictors, and outcome.

solid/cystic composition of the tumor. While PCP is a predominantly solid or mixed solid-cystic, spherical-shaped, located in a suprasellar location, the ACP frequently appears to be a lobulated cystic or predominantly cystic mass in an intrasellar/suprasellar location.[13] Differential diagnosis with other tumors such as germinoma, when located in the anterior skull base, is of importance, since it is an intracranial tumor sharing similar clinical manifestations and imaging features with CPs.[19] Clinical experts can diagnose pediatric suprasellar brain tumors, such as ACP, with an accuracy of up to 86%. However, the best accuracy is attained by surgical biopsy, reaching 91%. This is caused by limited interpretation of MRI or computed tomography scans due to the low spatial resolution of radiology images and atypical visual features that make diseases look similar.[12] Incomplete resection of CPs could be associated with poor prognosis. In this case, the histopathology of the interface between CPs and surrounding brain tissue represents finger-like local invasion, which shows the resection failure.[21] Therefore, optimizing preoperative noninvasive diagnostic tools should be enhanced to increase diagnostic accuracy and the successful choice of the treatment approach.[12,13,19,21] Radiomics, an emerging technique, obtains high-throughput features from digital imaging to quantitatively describe lesion characteristics.[11,20] These features show promising clinical application value, such as neuronavigation and prognostic analysis. With the aid of DL, ML algorithms, investigating the intrinsic characteristics of high-throughput image features is easier and more systematic.[11,20]

This systematic review and meta-analysis evaluated the effectiveness of AI in forecasting, segmentation, and classification of CPs. The main focus of included articles is segmentation and classification based on ML and DL algorithms. The results demonstrated that the pooled AUC for diagnostic performance was 0.91, sensitivity was 0.87, and specificity was 0.8. By diagnostic performance in this meta-analysis, the pooled AUC for classification performance was 0.899, and the pooled sensitivity for segmentation performance was 0.755. Outcomes show that AI's potential to aid in clinical decision-making and AI-based models have a promising performance in the diagnosis, segmentation, and classification of CPs.

X. Chen et al. systematically used segmentation and classification to obtain radiomic features regarding the distinction between ACP and PCP and specific genetic mutations of BRAF V600 E and CTNNB. Based on a three-stage feature selection method involving modification of the manually segmented tumor region, adopting an RF classification model, and finally, a sequential forward selection strategy, key features were selected. This study achieved strong prediction ability for ACP/PCP differentiation (0.89) and diagnosing BRAF V600 E (0.91) and CTNNB (0.93) mutations.[18] This brings a clinical impact since 2 patients with recurrent CPs with a BRAF V600 E mutation were subscribed to BRAF/MEK inhibitors. Target therapy worked well in a short period, resulting in a significant decrease in tumor size.[18] In a study by Y. Teng et al. among 6 segmentation-free CNN models, including VGG16, ResNet18, ResNet50, ResNet101, DenseNet121, and DenseNet169, the VGG16-based model and ResNet50-based model showed the best predictive performance, with AUC values of 0.822 and 0.838, respectively. Compared to ML-based radiomic models, with the best AUC achieved by the RF-based model (0.76), the CNN method had a competitive performance. CNN models using an automated end-to-end approach based on DL technology resolve the ML models' need for manual segmentation during feature extraction and further feature selection while achieving comparable sensitivity and specificity.[4] X. Yan et al. proposed a multistructure segmentation method for CP based on DL, which achieved an accuracy of 0.90, using sagittal MRI and suggests that the automatic segmentation and classification methods designed could be used as the primary method for identifying CPs rather than subjective judgments based on human experience. The proposed segmentation method can be used not only for three-dimensional reconstruction but also for intraoperative navigation. This classification is a notable multitissue segmentation standard that can display 6 adjacent morphological structures of CPs (tumors, pituitary gland, sphenoid sinus, brain, superior saddle cistern, and lateral ventricle) and established a multitissue automatic segmentation method for CPs, which achieved a segmentation Dice coefficient of 0.951 for tumors and an average Dice coefficient of 0.8526 for 6 tissues.[11] Automated tumor segmentation enhances both diagnostic evaluation and tumor growth monitoring. Automated segmentation provides an improved approach to routine clinical imaging assessment as it allows convenient detection, volumetric assessment, and precise therapy planning. Recent advances in radiomics also suggest that automated segmentation can provide an image preprocessing method for subsequent tumor analysis. Compared to laborious manual delineation, it is highly reproducible and alleviates clinicians' interrater and intrarater variability.[7] The study by Zh-S Huang et al. demonstrates the superiority of more advanced radiomic features compared to traditional clinical and conventional radiologic features. The clinic-radiological model, combining age and 3 conventional MRI features, showed a limited capacity to distinguish the pathological subtype of ACP from PCP (AUC = 0.67 in external validation set). Among 7 radiomic models based on SVM ML algorithm, the one combining T1-weighted (T1-w), T2-weighted, and contrast-enhanced T1-w showed the best performance among the 7 models, with AUC values of 0.899, 0.810, and 0.920 in the training, internal, and external validation cohorts, respectively.[13] A study by Ch. Jiang et al. compared the radiomics-based ML models with clinical practice in differentiation of the 4 most common types of cystic sellar lesions, namely Rathke's cleft cysts, cystic CPs, cystic pituitary adenomas, and pituitary apoplexy. The preoperative diagnostic accuracy in clinical practice is only 0.67, indicating ongoing challenges in the differential diagnosis of such conditions. Based on 6 ML methods, including RF, SVM, BaggingClassifier-SVM, AdaBoost, AdaBoost_DT, and LR paired imaging differentiations (i.e., Rathke vs. CP) were performed, which resulted in an average AUC value of 0.76 and accuracy of 0.75. Regarding accuracy, the ML models outperformed the clinical knowledge-based model by approximately 8%.[20] Similarly, in one approach to use radiomics-based ML models to differentiate between germinoma and CPs, based on T1-w images, three classification algorithms were used, including linear discriminant analysis, SVM, and RFC. Nine models were established, trained, and validated with different selection methods and classifier combinations. The results of this study showed that RFS (RF feature selector) constructed the best prediction models with RFC and LASSO with linear discriminant analysis in the contrast-enhanced T1WI, observed to be the optimal methods with AUCs above 0.9.[19]

Technical Considerations

Interestingly, the pooled classification performance for ACP versus PCP (AUC = 0.899) still exceeds overall diagnostic performance, with the AUC for diagnosis now improving to 0.91. This enhancement reflects a more robust diagnostic capability for AI models. The improved performance is likely due to the more homogeneous datasets and consistent imaging protocols, allowing AI to more accurately differentiate CPs from other lesions in the sellar and parasellar regions. Furthermore, the higher classification accuracy (AUC = 0.899) can be attributed to the distinct radiomic signatures between ACP and PCP, which AI models can identify with high fidelity.

Limitation

Most studies relied on retrospective, single-center designs, which inherently invite bias and can result in substantial missing data. Several lacked clearly defined training versus testing cohorts, an essential element in AI research, making it impossible to derive TP, TN, FP, and FN rates. Small sample sizes and brief follow-up periods likely underrepresented patients with poorer outcomes. Furthermore, over 70% of the included studies were performed in China, raising selection bias concerns and limiting the broader applicability of our findings. Finally, the absence of reported standard errors or CI in several studies forced their exclusion from our quantitative meta-analysis.

Future Directions

To enhance the usability and effectiveness of ML models for diagnosing, segmenting, and classifying CPs, future primary studies should enlarge their datasets through prospective, multicenter collaborations that include more numerous and varied patient cohorts. It is vital to employ rigorous study designs with clearly separated training and testing sets, sufficient sample sizes, and extended follow-up to ensure precise estimation of TP, TN, FP, and FN. Harmonizing imaging protocols, particularly MRI acquisition parameters, will further reduce variability. Researchers should also refine and compare different ML techniques, including DL architectures, to boost predictive performance across CP subtypes. Lastly, externally validating these models in independent multicenter cohorts is essential to confirm their generalizability and real-world applicability.

CONCLUSION

This study highlights the transformative potential of AI in the diagnosis, segmentation, and classification of CPs. By leveraging ML and DL, AI not only enhances the diagnostic accuracy but also streamlines clinical decision-making processes. For radiologists and neurosurgeons, these AI-driven tools offer powerful support in overcoming the challenges of complex tumor identification and preoperative planning. As AI models continue to evolve, their integration into routine clinical practice promises to revolutionize how CPs are managed, leading to more precise, personalized, and effective treatment strategies. The future of AI in neuro-oncology is not just about improving outcomes but about reshaping the entire landscape of tumor diagnosis and management.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Ibrahim Mohammadzadeh: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Validation, Writing — original draft, Writing — review & editing. **Bardia Hajikarimloo:** Formal analysis, Methodology, Writing — original draft. **Behnaz Niroomand:** Data curation, Formal analysis, Methodology, Writing — original draft. **Nasira Faizi:** Data curation, Validation. **Nasiha Faizi:** Data curation, Validation. **Mohammad Amin Habibi:** Resources, Writing — review & editing. **Shahin Mohammadzadeh:** Investigation, Supervision, Writing — review & editing. **Reza Soltani:** Writing — review & editing.

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